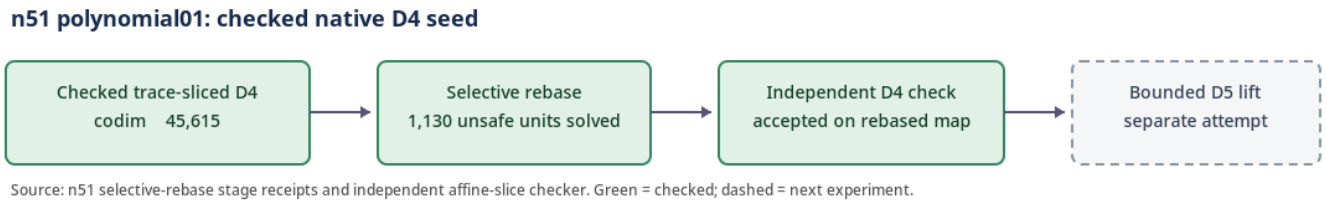
A checked native D4 seed for the n51 degree-five lift



Source: n51 selective-rebase stage receipts and independent affine-slice checker. Green = checked; dashed = next experiment.

The named **fresh-n51-poly-01** trace-sliced D4 map was rebased into the native unit-triangular form. An independent checker accepted all C0-C3 conditions on 51 sliced generators, including a nonzero image of 1. Its codimension lower bound is **45,615** and low-kernel dimension is 10,194.

Stage	Recorded result	Frozen limit
Unit layout	44,485 safe; 1,130 unsafe literal units	Read-only
Selective rebase	Exit 0; 45,615 unit representatives	8 GiB; 1,800 s
Independent D4 check	Accepted; codim at least 45,615	16 GiB; 3,600 s

The 1,553,961,056-byte rebased map has SHA-256 `be38aff828d9d3c73e70ea78914727fdca1305edbf5f99e99b28aa747cc396d8`. The native rebase retained the safe units and chose 1,130 other monomial representatives. Small adversarial and multiword fixtures checked full kernel preservation; the independent checker then verified the actual named equations and trace slice.

The read-only D5 profile gives 58,262 border and 103,877 working coordinates. Its 4096-row source-derived allocation envelope is 55,355,297,148 bytes. This is a sizing calculation, not measured RSS or a D5 result. A separately frozen D5 attempt needs a fresh exact original count and any candidate must pass independent D5 checking.

The USD 4 reservation, exact source and binary digests, accepted checker transcript, and replay commands are in report.md, evidence.json, and raw/ alongside this PDF.